Case Study: Dynamic UPI Transaction Analytics Dashboard

An end-to-end data analysis project transforming over 500,000 raw financial transaction records into an interactive executive dashboard, uncovering critical payment performance insights and failure trends.

1. Executive Summary & Objective

As Unified Payments Interface (UPI) transaction volumes grow exponentially, payment service providers require real-time visibility into transaction health, app market share, and failure patterns. The objective of this project was to analyze a high-volume UPI transaction dataset and build a dynamic, self-service executive dashboard in Microsoft Excel. The resulting tool enables stakeholders to monitor key performance indicators (KPIs), evaluate payment app performance, and identify operational bottlenecks without relying on expensive BI infrastructure.

2. Data Scope & Analyst Actions

The analysis evaluated a dataset of over **500,000 transaction records** across major UPI applications (PhonePe, Google Pay, Paytm, BHIM, Amazon Pay) with amounts standardized in Indian Rupees (INR ₹). Key actions taken during the analysis process included:

- **Data Cleaning & Formatting:** Standardized raw timestamps, resolved date syntax errors, and eliminated null/blank entries to establish a clean source dataset.
- **Feature Engineering:** Extracted temporal dimension attributes (e.g., Hour of Day, Day of Week) to enable granular time-series and peak-traffic evaluation.
- **Performance Optimization:** Optimized calculation pipelines and flattened intermediate data structures to support high-speed interactive filtering across half a million rows.

3. Dashboard Architecture

To maintain clean UI separation and maintainability, the solution was constructed using a structured three-tier architecture:

Layer	Sheet Name	Function & Responsibilities
Backend	Raw_Data	Stores static transaction logs. Includes helper columns (Hour, Day_Name) calculated once and flattened to preserve system resources.

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Middleware	Calc_KPI / Pivots	Houses underlying Pivot Tables, dynamic aggregation structures, and helper logic using standard data retrieval formulas.
Frontend	Dashboard	Highly polished, non-grid presentation layer featuring dynamic KPIs, interactive slicers, charts, and heatmaps.

4. Key Analytical Insights Uncovered

Payment Performance & Failure Rates

- **Failure Rate Disparities:** Transaction success rates varied noticeably across payment apps, identifying specific platforms with elevated technical decline rates during peak usage.
- **Temporal Peak Stress Points:** Hour-by-hour matrix analysis revealed concentrated traffic bursts during evening hours (18:00–21:00), correlating directly with increased transaction latency and status drop-offs.

App Ecosystem & Market Share Trends

- **Volume Distribution:** Market share analysis highlighted heavy transaction concentration among top-tier apps, while smaller apps showed higher average order values (AOV).
- **Demographic & Regional Patterns:** Transaction volume distributions underscored key high-density geographic corridors driving the majority of financial throughput.

Business Value & Strategic Recommendations

- **Proactive Server Load Balancing:** Recommended infrastructure capacity adjustments during identified peak window traffic to reduce technical decline rates.
- **Targeted App Partnership Focus:** Identified partner apps requiring technical routing support to improve transaction completion rates.
- **Executive Decision Support:** Delivered an intuitive, lightweight solution that empowers leadership to make data-driven decisions rapidly without SQL or BI tool overhead.